Courier Management System(SMS)

Team ID : SWTID-2026-3928

Team Size : 5

Team Leader : SASIKUMAR V

Team member : IYYANAR A

Team member : DINESH P

Team member : SURYA PRAKASH K

Team member : VIGNESH S

ABSTRACT

The Courier Management System is a software application designed to manage and track courier services efficiently. It helps courier companies handle parcel booking, customer details, shipment tracking, and delivery status in an organized way. The system reduces manual work and improves accuracy in managing courier operations.

In this system, users can enter sender and receiver details, parcel information, and delivery location. The system stores all the data in a database and allows easy tracking of courier status such as dispatched, in transit, and delivered. It also helps administrators manage records, update delivery status, and generate reports.

The main objective of this project is to make courier service management faster, more reliable, and user-friendly. By using this system, courier companies can improve service quality and reduce errors in data management.

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1. INTRODUCTION

The Courier Management System is a software application used to manage courier services in an efficient and organized way. In courier companies, many parcels are sent and received every day. Managing all these records manually can be difficult and time-consuming. To solve this problem, a computerized system is used.

This system helps to store and manage information such as sender details, receiver details, parcel information, and delivery status. It also allows the user to track the courier and update its status easily. By using this system, the chances of errors in manual record keeping are reduced.

The main purpose of the Courier Management System is to make courier operations faster, safer, and more reliable. It helps courier companies improve their service and manage their data effectively

2. IDEATION PHASE

The Ideation Phase is the stage where ideas for the Courier Management System are generated and planned. In this phase, the main problems in the existing courier process are identified and possible solutions are discussed.

Many courier services manage parcel details manually, which can lead to errors, delays, and difficulty in tracking shipments. To overcome these issues, the idea of developing a Courier Management System is proposed. This system helps to manage sender and receiver details, parcel information, delivery status, and tracking in a digital way.

During this phase, the main features of the system are planned, such as parcel booking, shipment tracking, updating delivery status, and

managing customer records. The goal is to design a system that makes courier operations faster, more accurate, and easy to manage.

3 REQUIREMENT ANALYSIS PHASE

The Requirements Analysis Phase is the stage where the needs and requirements of the Courier Management System are identified and analyzed. In this phase, the system requirements are collected to understand what features and functions the system should provide.

The system must be able to store and manage important information such as sender details, receiver details, parcel information, and delivery status. It should allow users to add, update, delete, and view courier records easily. The system should also provide a tracking facility to check the current status of the parcel.

In this phase, both functional requirements (what the system should do) and non-functional requirements (system performance, reliability, and security) are considered. The main objective of this phase is to clearly define the system requirements so that the development process can be carried out effectively.

Important Functional Requirements

These are the main functions the system must perform.

- User Login – Admin or staff can login to the system securely.
- Add Courier Details – Enter sender name, receiver name, address, and parcel details.
- Update Courier Status – Update status like dispatched, in transit, delivered.

- Track Courier – Users can track the parcel using a tracking number.
- Manage Customer Records – Add, edit, or delete customer information.
- View Courier Details – View all courier records stored in the database.

Non-Key (Non-Functional) Requirements

These describe how the system should work, not the main functions.

- Performance – The system should respond quickly when searching or updating data.
- Security – Only authorized users should access the system.
- Reliability – The system should work without errors or data loss.
- User-Friendly Interface – The system should be easy to use.

4. PROJECT DESIGN PHASE

The Project Development Phase is the stage where the actual coding and creation of the Courier Management System takes place. In this phase, the system is developed based on the design and requirements defined earlier.

The developers create different modules such as user login, courier booking, customer details, parcel tracking, and delivery status update. The database is also created to store information like sender details, receiver details, and parcel records.

Programming languages and tools are used to build the system interface and connect it with the database. Each module is developed and integrated to make the system work properly.

5.PROJECT PLANNING & SCHEDULING

The Project Planning and Scheduling Phase is the stage where the activities and timeline of the project are planned. In this phase, the project tasks are organized and scheduled to complete the project within the given time.

The project is divided into different phases such as requirement analysis, system design, development, testing, and implementation. Each phase is given a specific time period for completion.

Planning helps the team understand what work should be done, who will do it, and when it should be completed. Scheduling ensures that all tasks are completed in an orderly manner without delay.

The main objective of this phase is to manage the project efficiently and complete the Courier Management System on time.

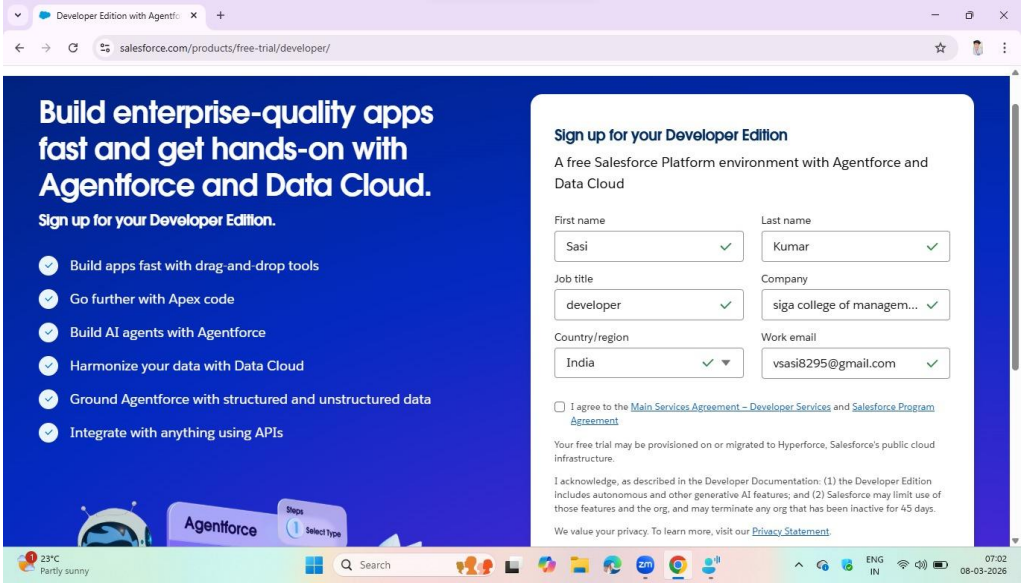
6. Project Development Phase:

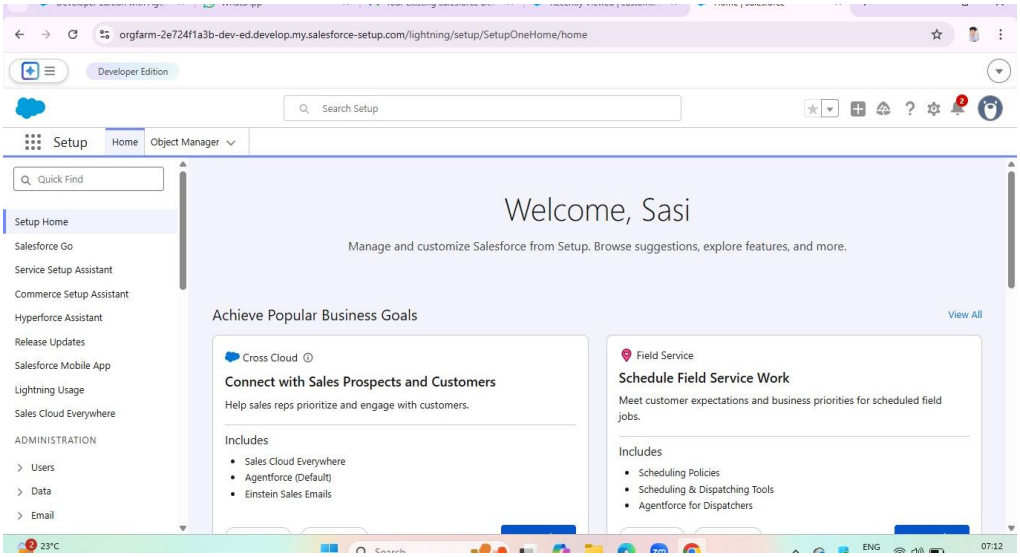
What you'll learn

1. Real-time Salesforce project implementation
2. Designing custom objects and relationships
3. Creating page layouts & record types
4. Writing validation rules
5. Building automation with flows
6. Developing reports & dashboards
7. Setting up security with profiles, permission sets, and sharing
8. End-to-end application testing

Milestone 1: Salesforce Account

1.1 Creating Developer Account & Account Activation





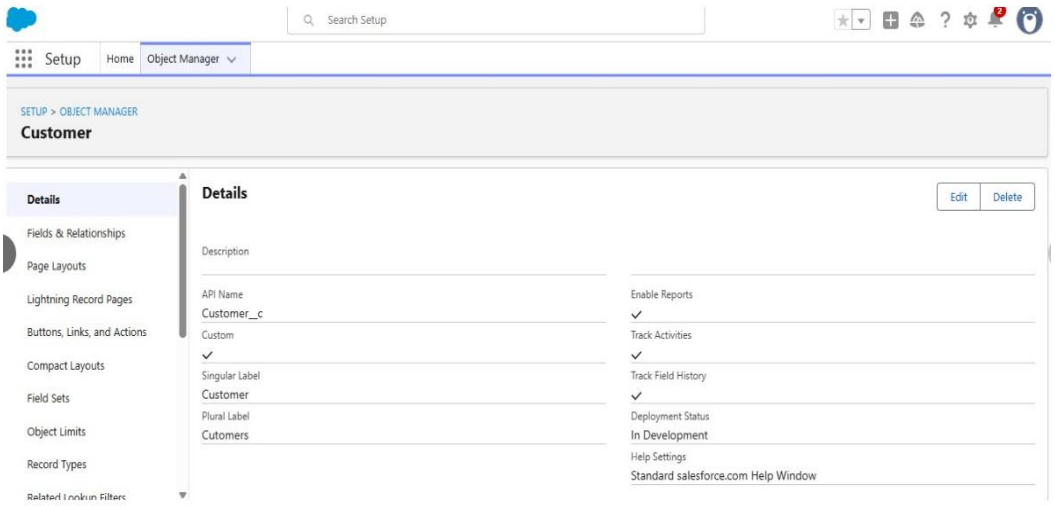
Milestone 2- Object:

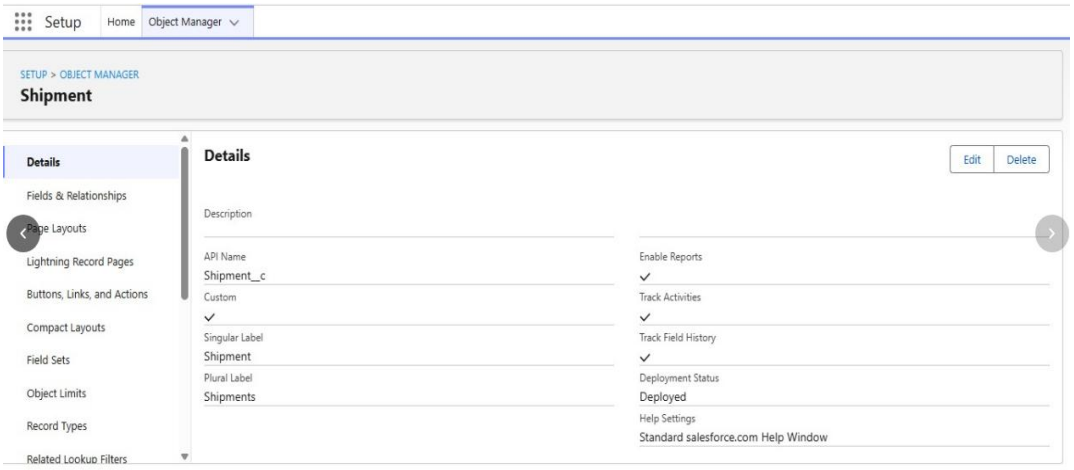
Object:

An object in Salesforce is a database table used to store specific information with fields and records.

Steps to Implement :

Create a (Customer, Shipment, Delivery_Agent, Branch, Delivery_Update, Invoice Object).





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SETUP > OBJECT MANAGER

Delivery Agent

- Details
- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters

Details

Edit | Delete

Description

API Name: Delivery_Agent__c

Custom: ✓

Singular Label: Delivery Agent

Plural Label: Delivery Agents

Enable Reports: ✓

Track Activities: ✓

Track Field History: ✓

Deployment Status: Deployed

Help Settings: Standard salesforce.com Help Window

Search Setup

Setup | Home | Object Manager

SETUP > OBJECT MANAGER

Branch

- Details
- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters

Details

Edit | Delete

Description

API Name: Branch__c

Custom: ✓

Singular Label: Branch

Plural Label: Branches

Enable Reports: ✓

Track Activities: ✓

Track Field History: ✓

Deployment Status: Deployed

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Search Setup

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SETUP > OBJECT MANAGER

Delivery Update

- Details
- Fields & Relationships
- Page Layouts
- Lightning Record Pages
- Buttons, Links, and Actions
- Compact Layouts
- Field Sets
- Object Limits
- Record Types
- Related Lookup Filters

Details

Edit | Delete

Description

API Name: Delivery_Update__c

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Singular Label: Delivery Update

Plural Label: Delivery Updates

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Track Field History: ✓

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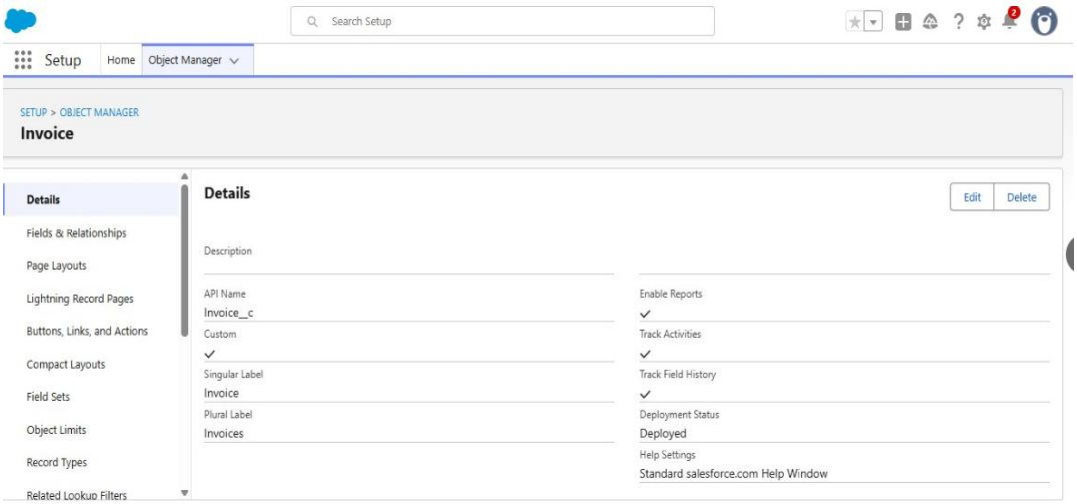
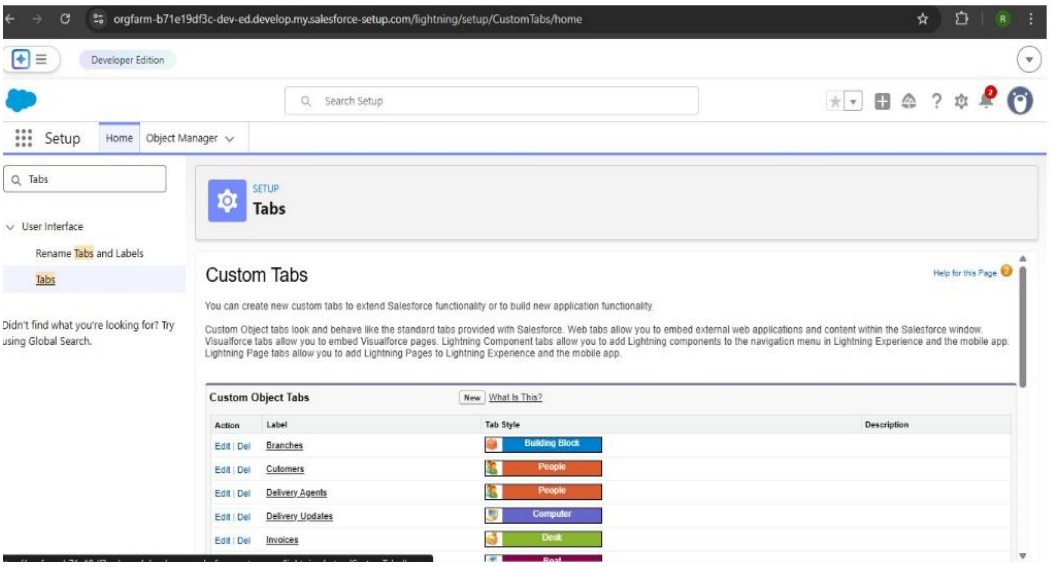


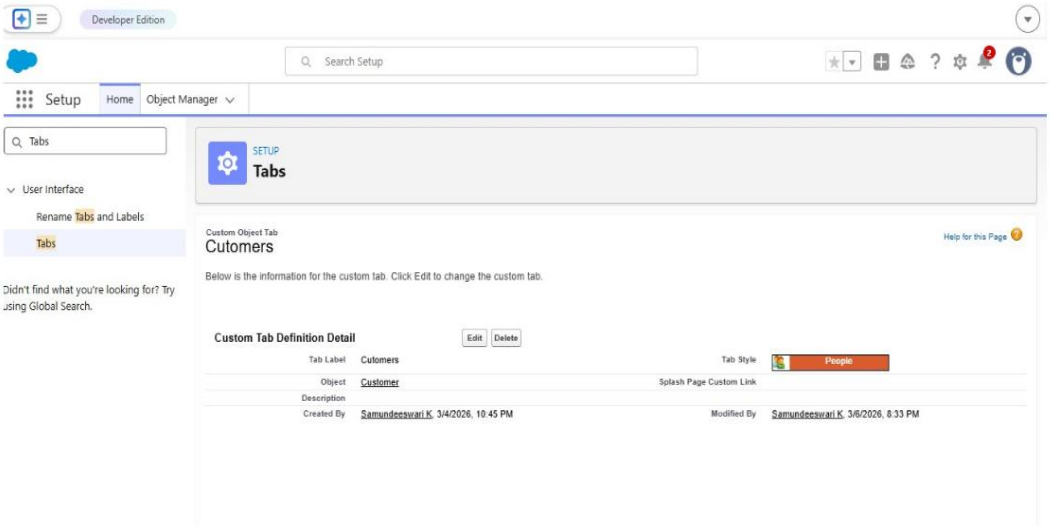
Milestone 3 - Tabs:

Tab:

Tabs in Salesforce are user interface elements that help users easily access and view data stored in objects.

Creating a Tab for (Customer, Shipment, Delivery_Agent, Branch, Delivery_Update, Invoice Object).

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Below is the information for the custom tab. Click Edit to change the custom tab.
Custom Tab Definition Detail
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Tab Label
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Tab Style
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Object
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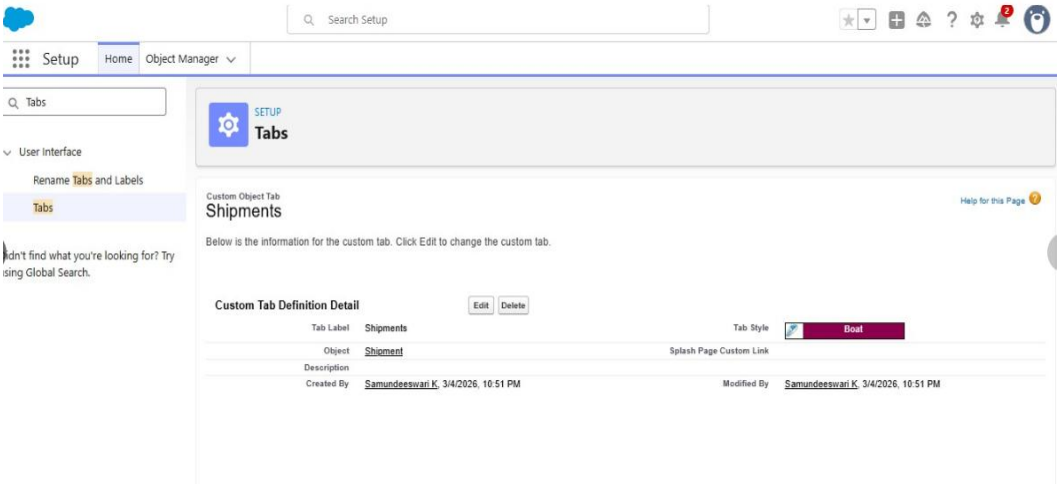
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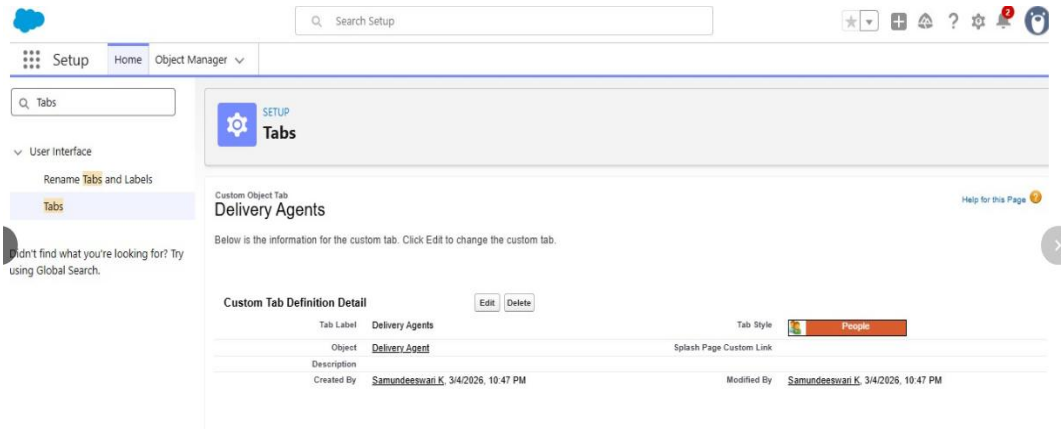
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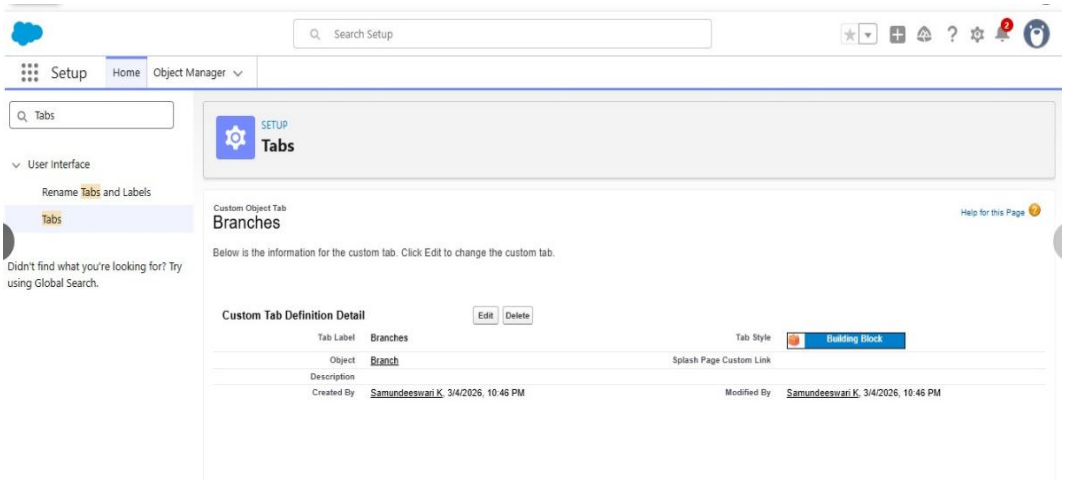
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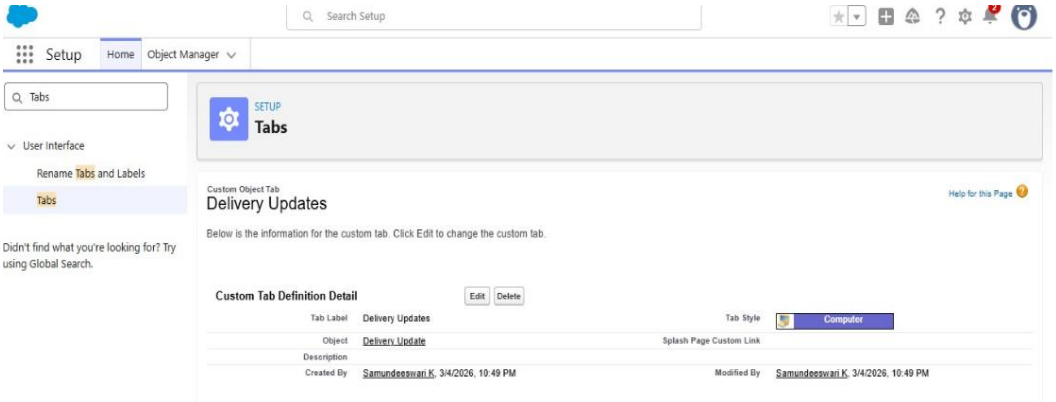
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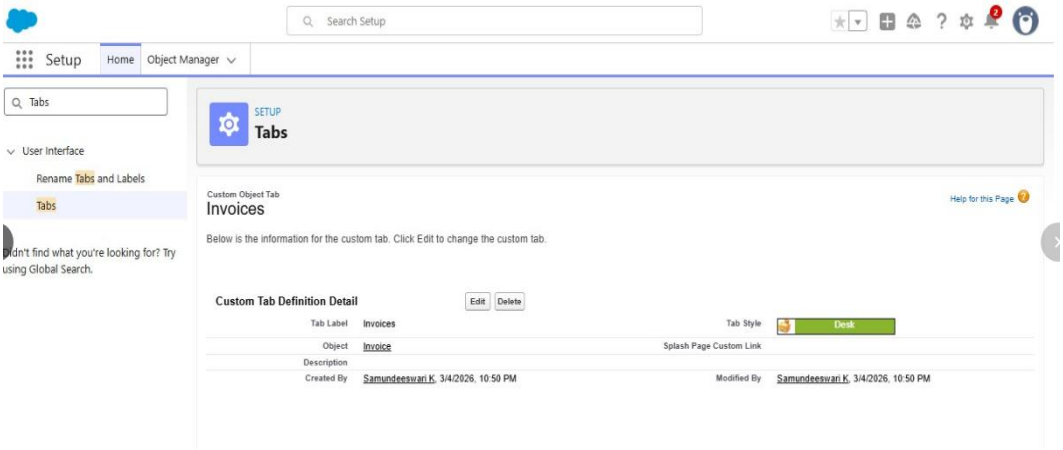
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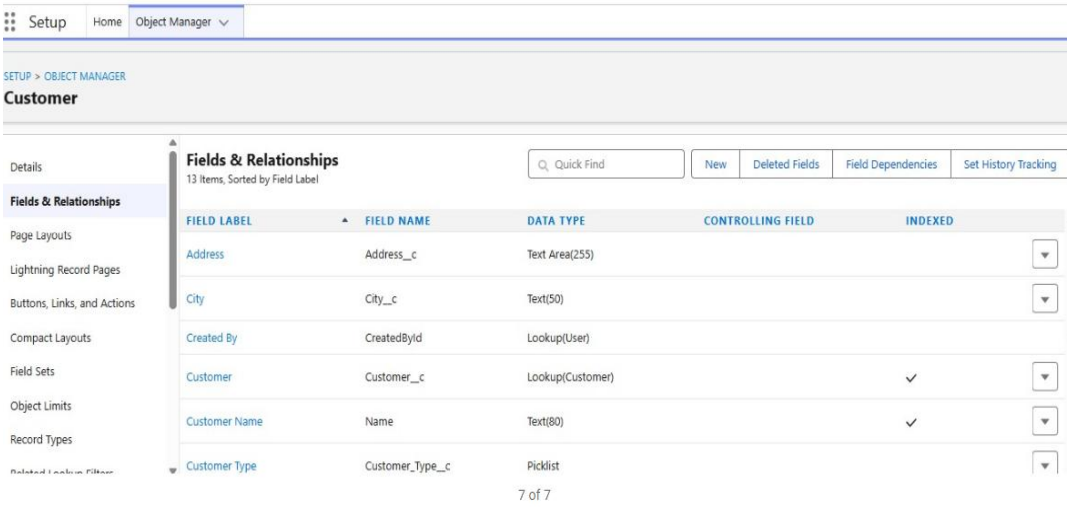
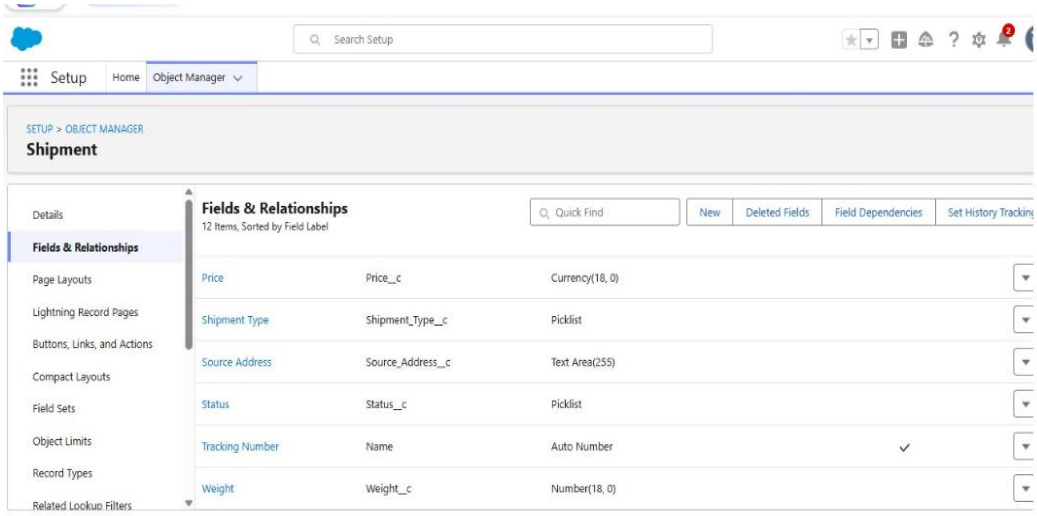
Milestone 4: Fields & Relationships

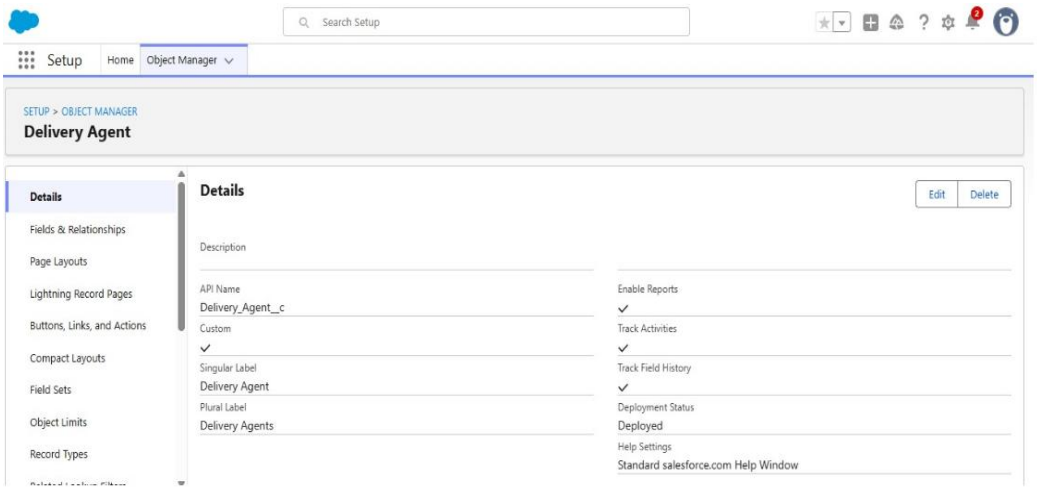
Fields:

A field in Salesforce is used to store a particular piece of data in a record.

Creation of fields for the (Customer,Shipment, Delivery_Agent, Branch, Delivery_Update, Invoice Object).



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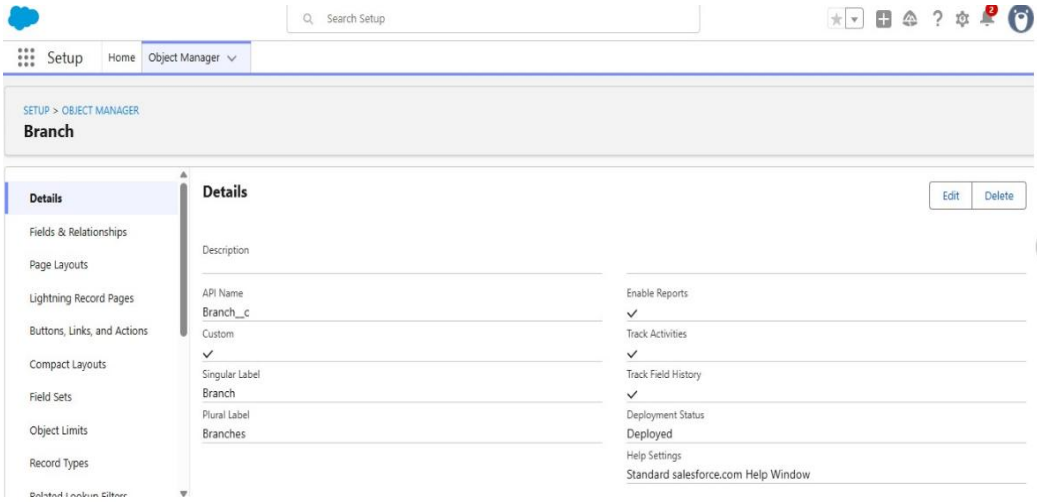
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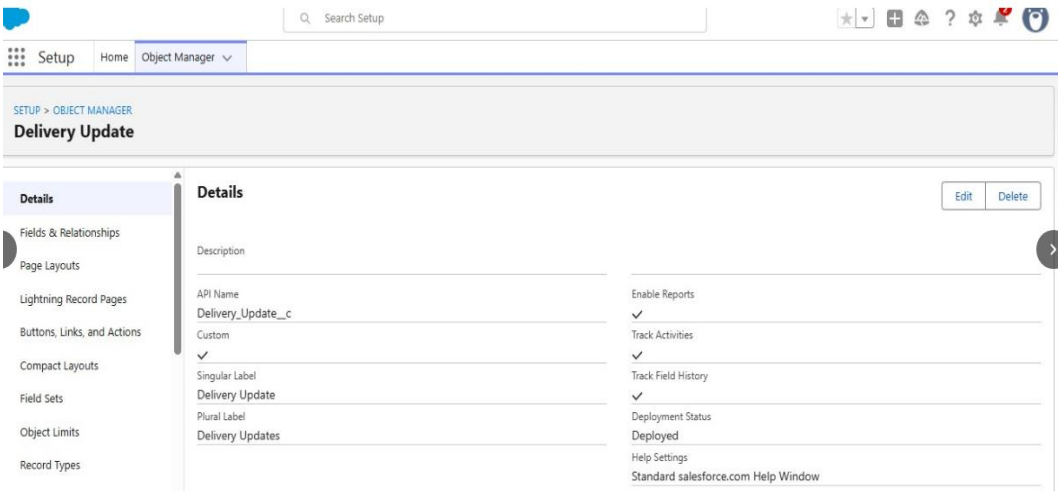
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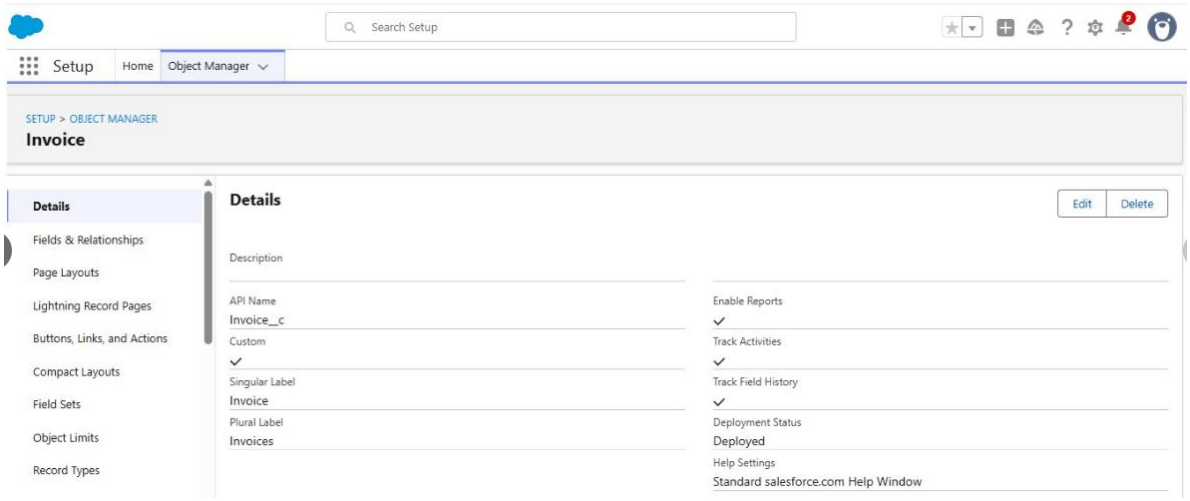
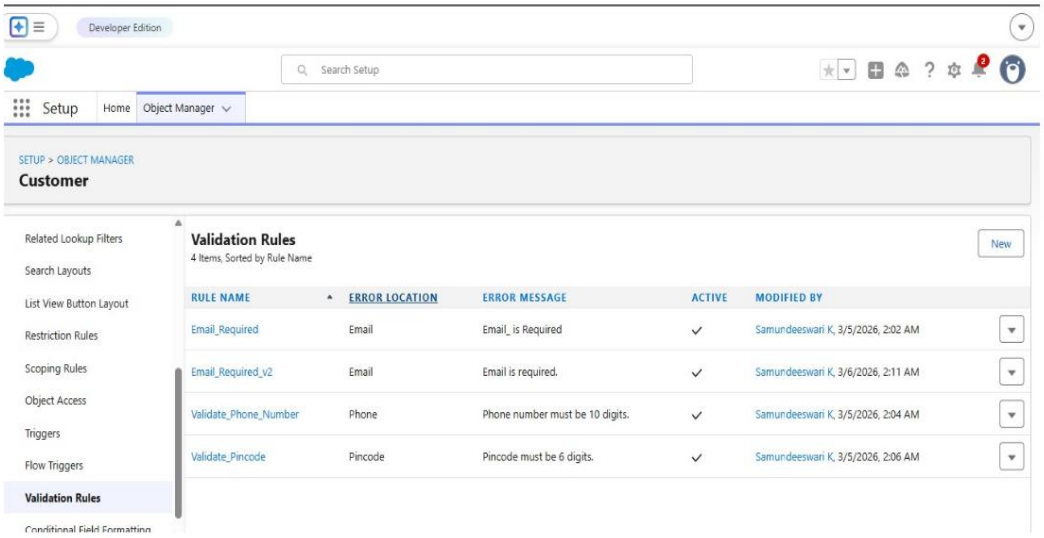


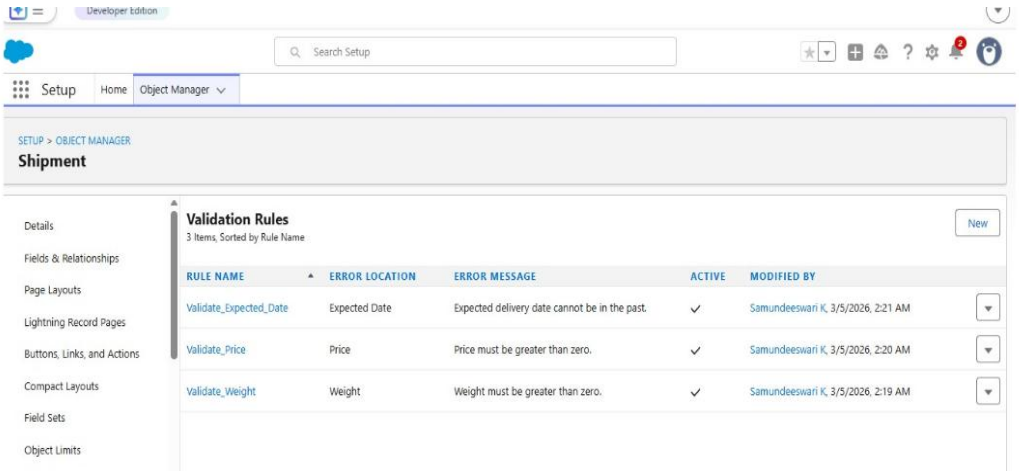
Milestone 5-Validation Rules :

Validation Rules:

A rule that checks whether the data entered in a field is correct before saving the record.

Create Validation Rules on (Customer_c,Shipment_c, Delivery_Agent_c, Branch_c, Delivery_Update_c).

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Object Limits
Validation Rules
3 Items, Sorted by Rule Name
New
RULE NAME
ERROR LOCATION
ERROR MESSAGE
ACTIVE
MODIFIED BY
Validate_Expected_Date
Expected Date
Expected delivery date cannot be in the past.
Samundeeswari K, 3/5/2026, 2:21 AM
Validate_Price
Price
Price must be greater than zero.
Samundeeswari K, 3/5/2026, 2:20 AM
Validate_Weight
Weight
Weight must be greater than zero.
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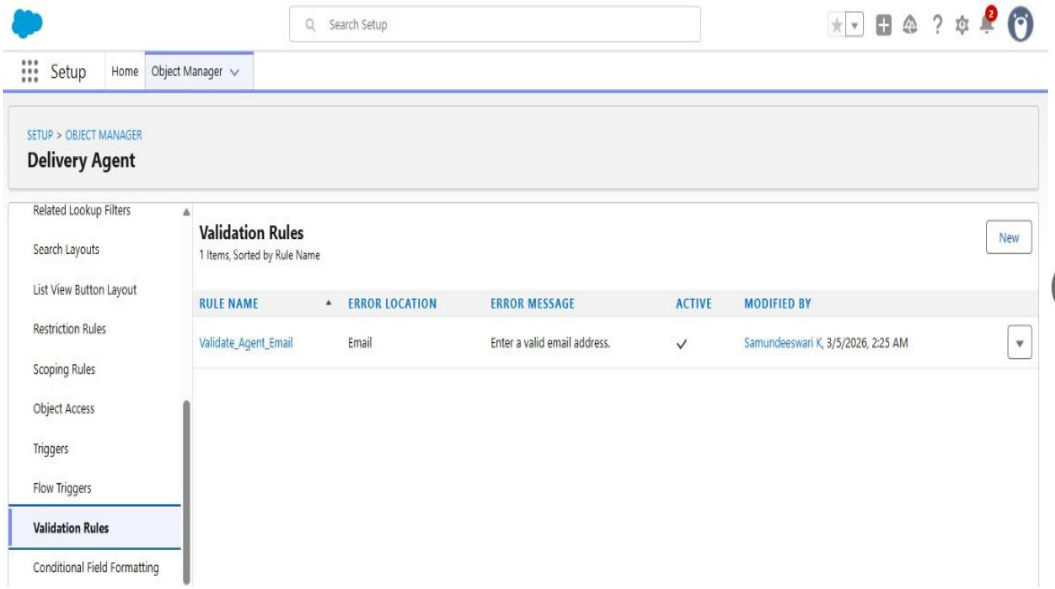
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Search Layouts
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Triggers
Flow Triggers
Validation Rules
Conditional Field Formatting
Validation Rules
1 Items, Sorted by Rule Name
New
RULE NAME
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ERROR MESSAGE
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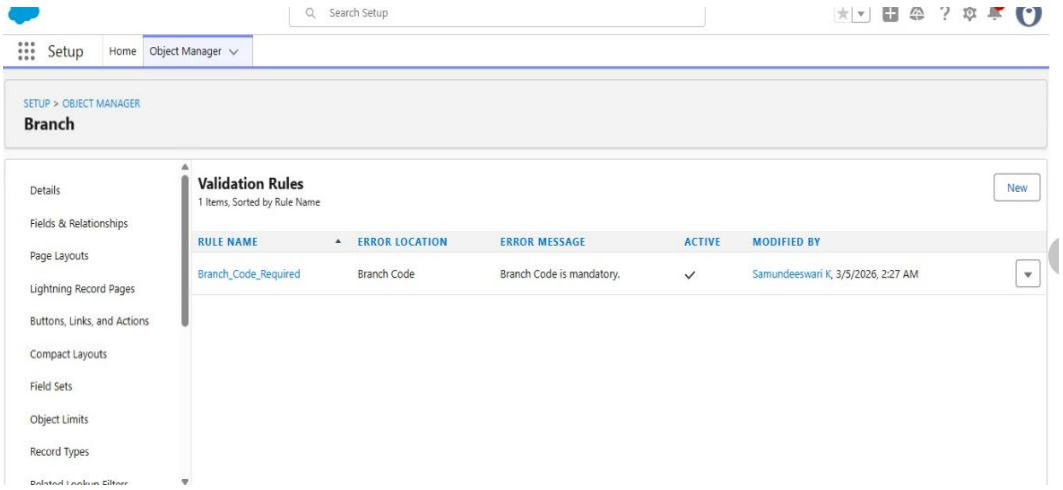
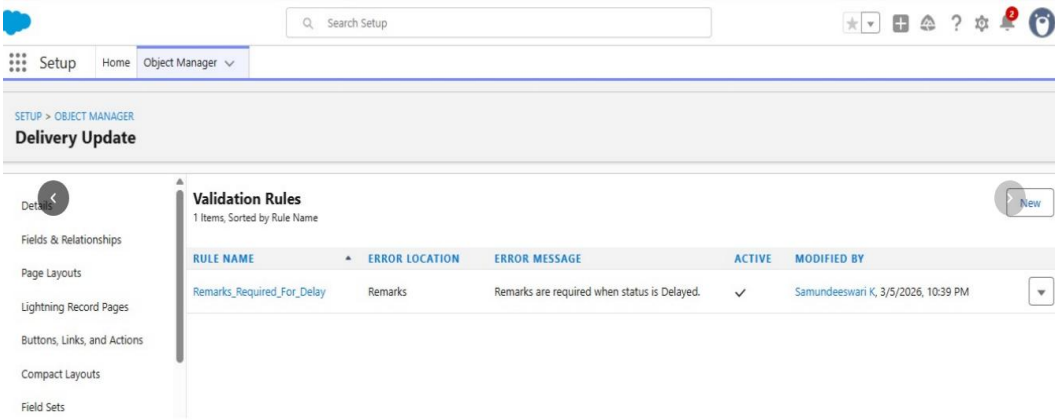


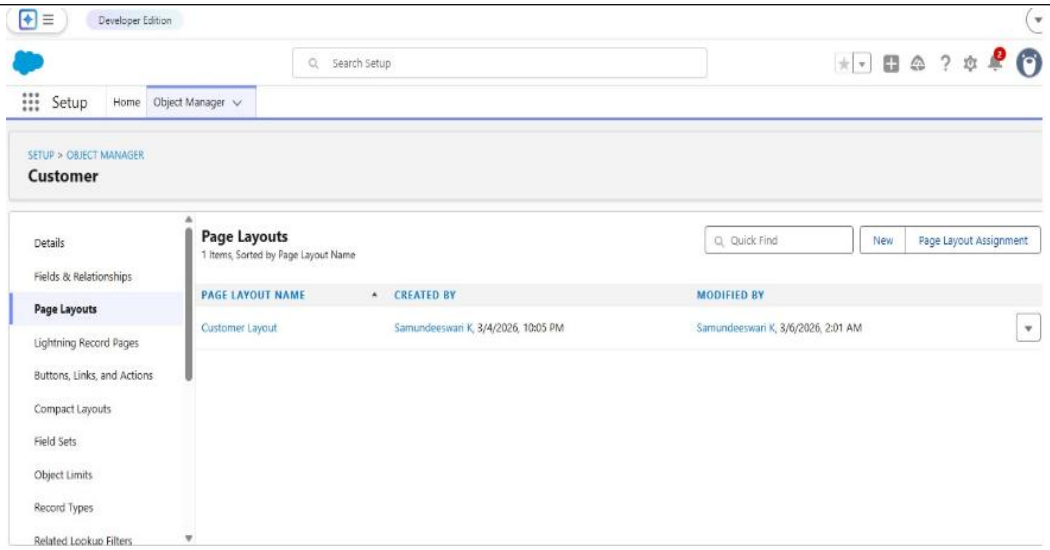


Milestone 6 - Page Layouts :

Page Layout

Page Layout is used to control how fields, buttons, and related lists appear on a record page.

Edit Page Layout for (Customer__c,Shipment_c, Delivery_Agent_c, Branch_c, Delivery_Update_c,Invoice_c).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Field Sets
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Record Types
Related Lookup Filters
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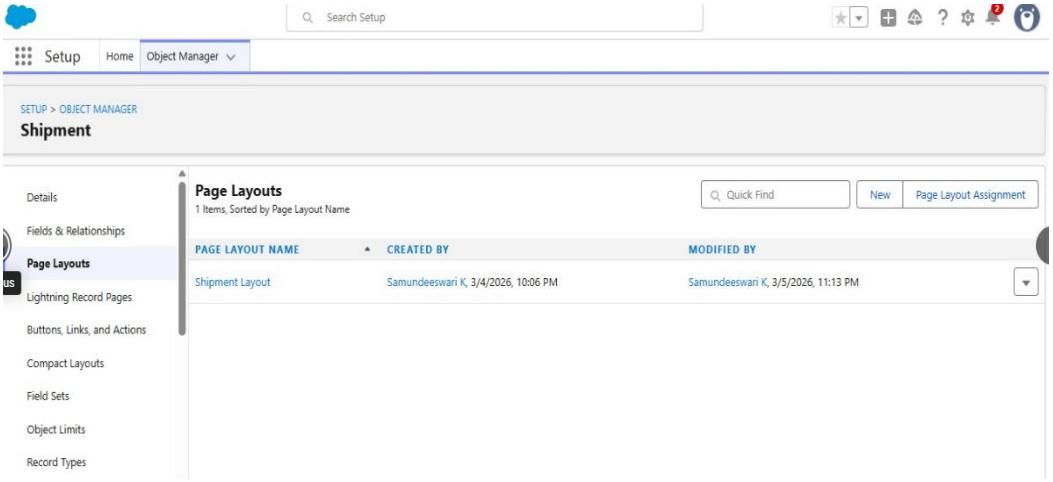
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Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Page Layouts
1 Items, Sorted by Page Layout Name
Quick Find
New
Page Layout Assignment
PAGE LAYOUT NAME
CREATED BY
MODIFIED BY
Shipment Layout
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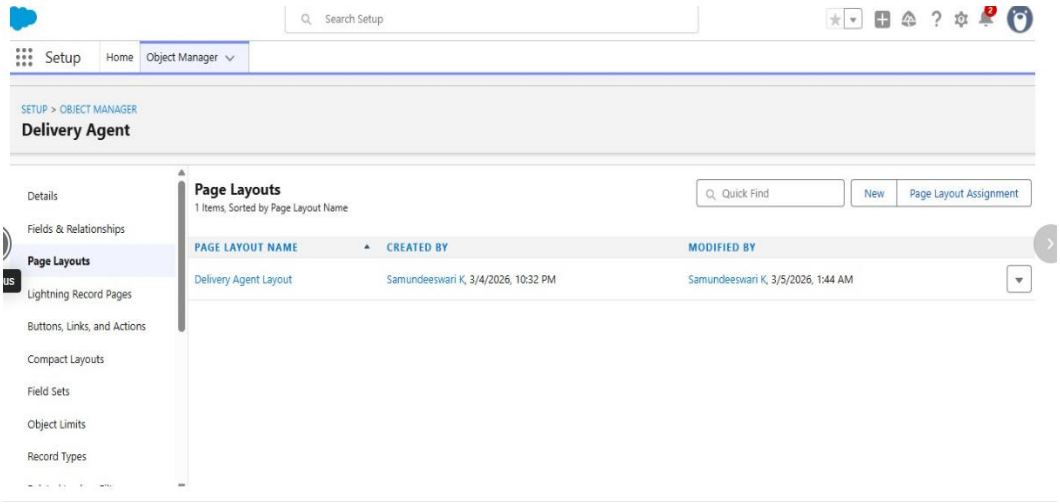
Samundeeswari K, 3/5/2026, 11:13 PM


Search Setup
Setup
Home
Object Manager
SETUP > OBJECT MANAGER
Delivery Agent
Details
Fields & Relationships
Page Layouts
Lightning Record Pages
Buttons, Links, and Actions
Compact Layouts
Field Sets
Object Limits
Record Types
Page Layouts
1 Items, Sorted by Page Layout Name
Quick Find
New
Page Layout Assignment
PAGE LAYOUT NAME
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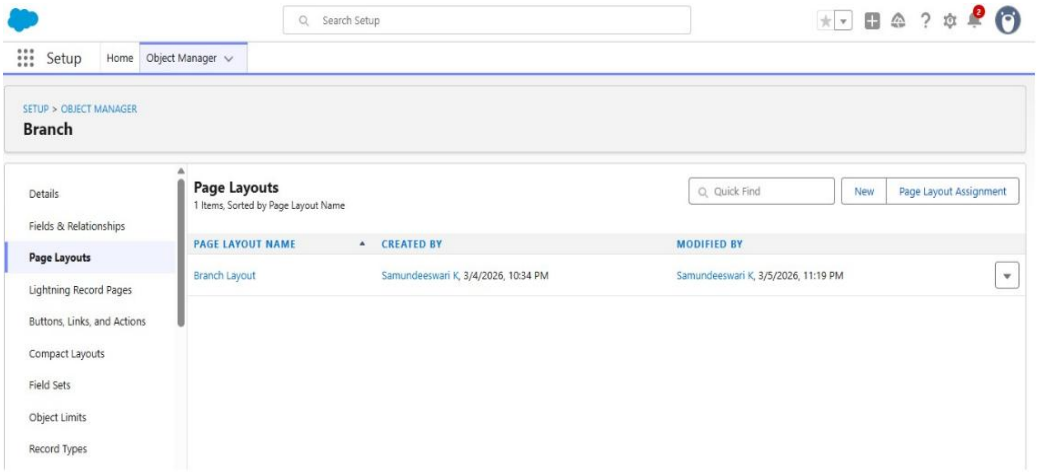
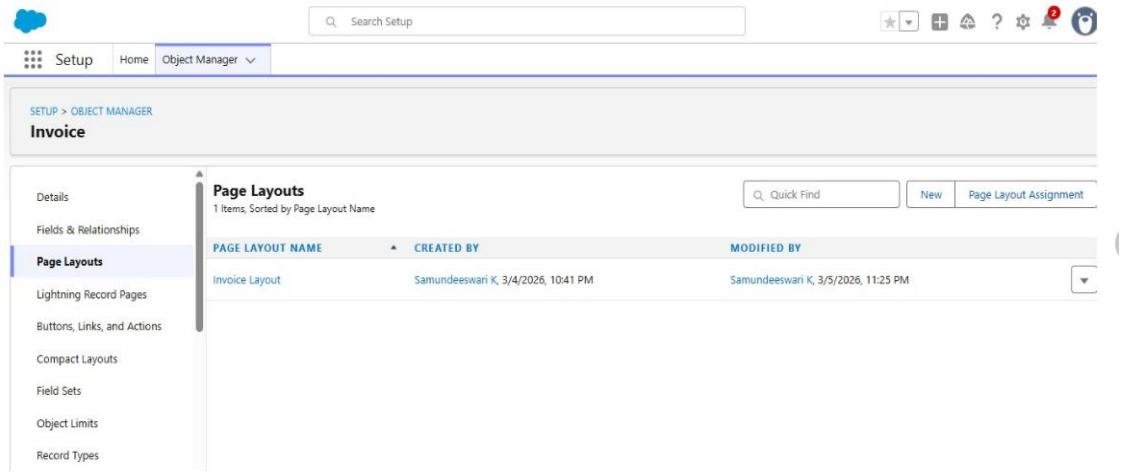
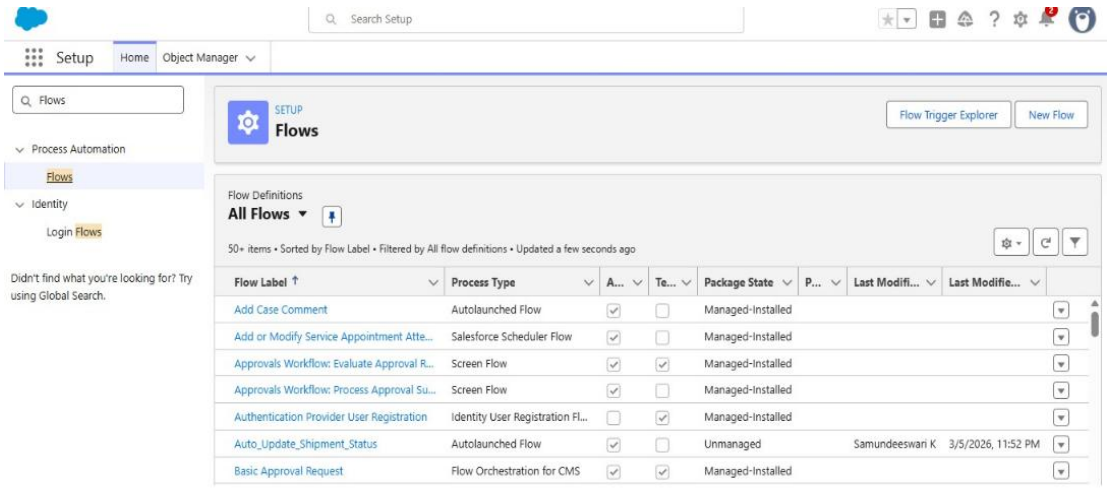




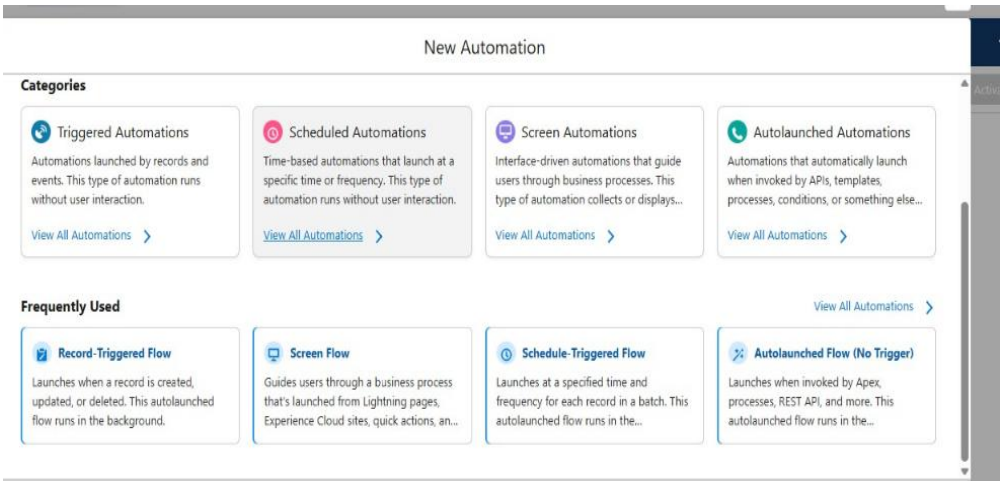
Milestone 7 - Flows

Flows

Flow is used to automate business processes by collecting data and performing actions automatically.



Create the Record-Triggered Flow

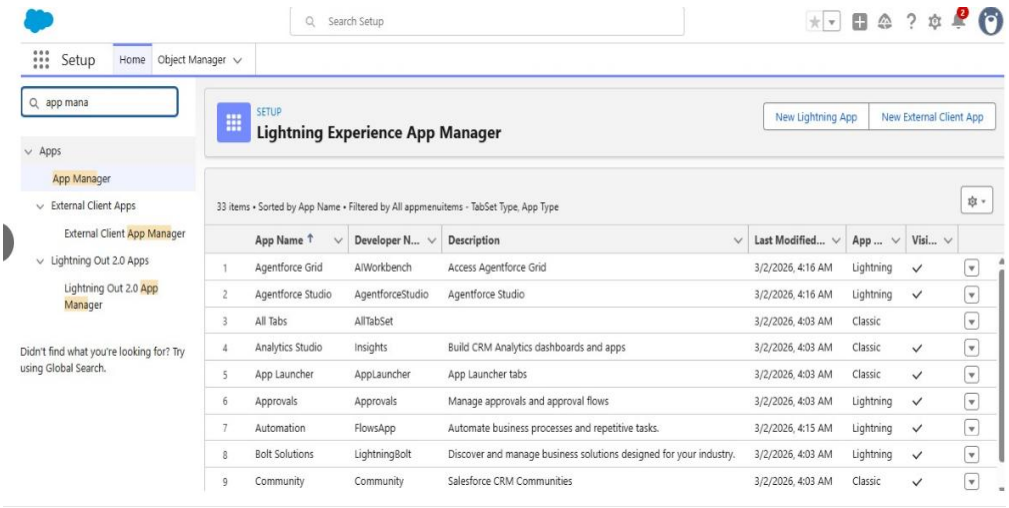


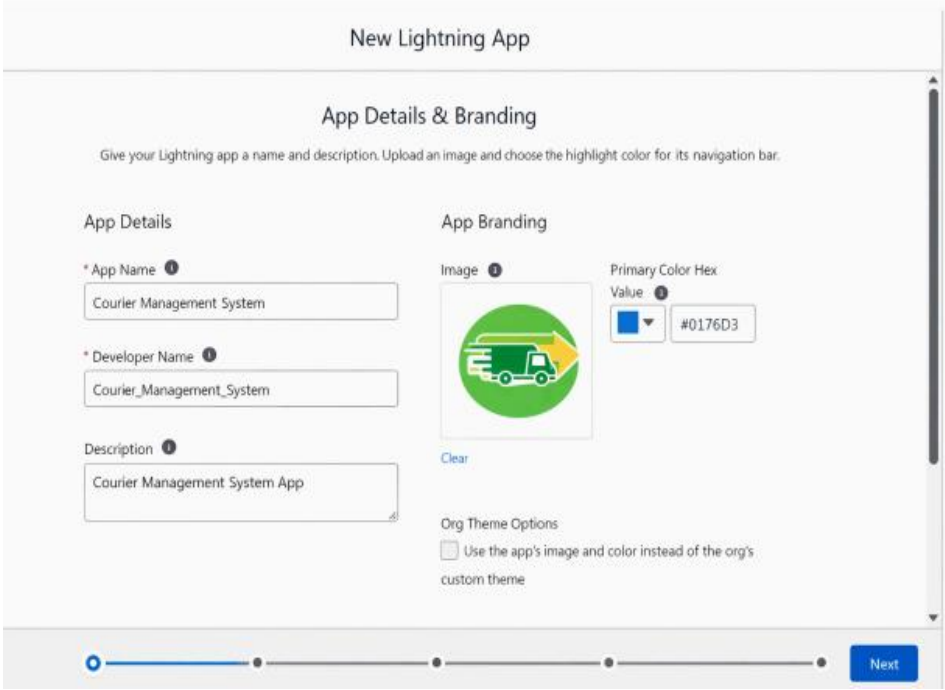
Milestone 8 - The Lightning App :

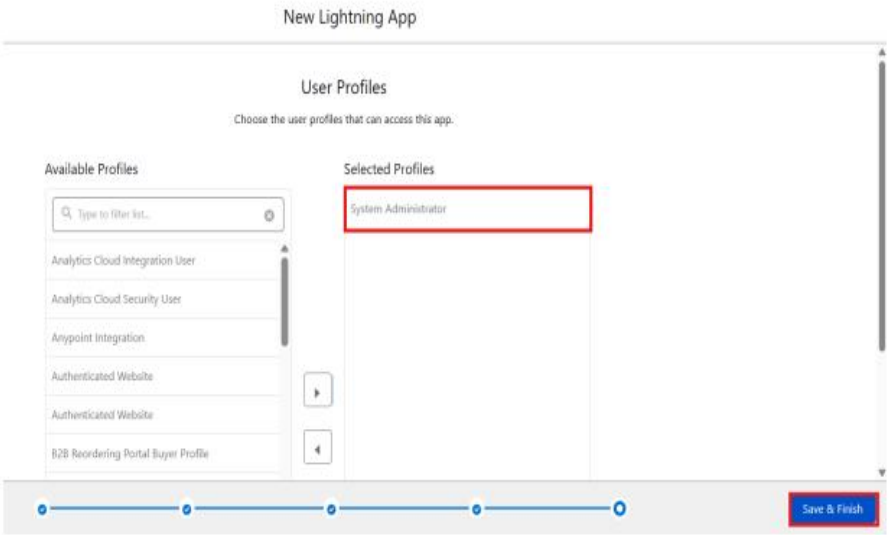
Lightning App

Lightning App is an application in Salesforce that groups related objects, tabs, and components into a single workspace for users to perform tasks easily.

Create the Courier Management System App:





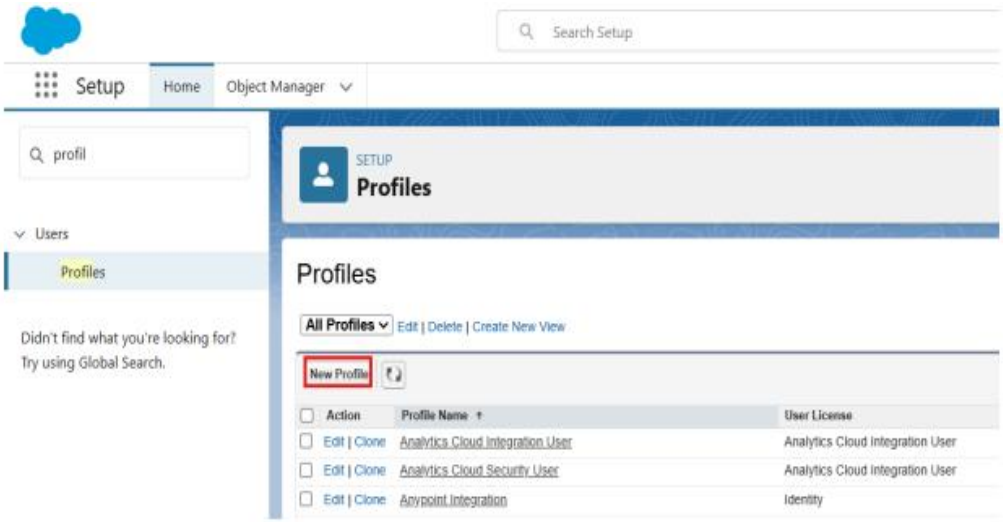


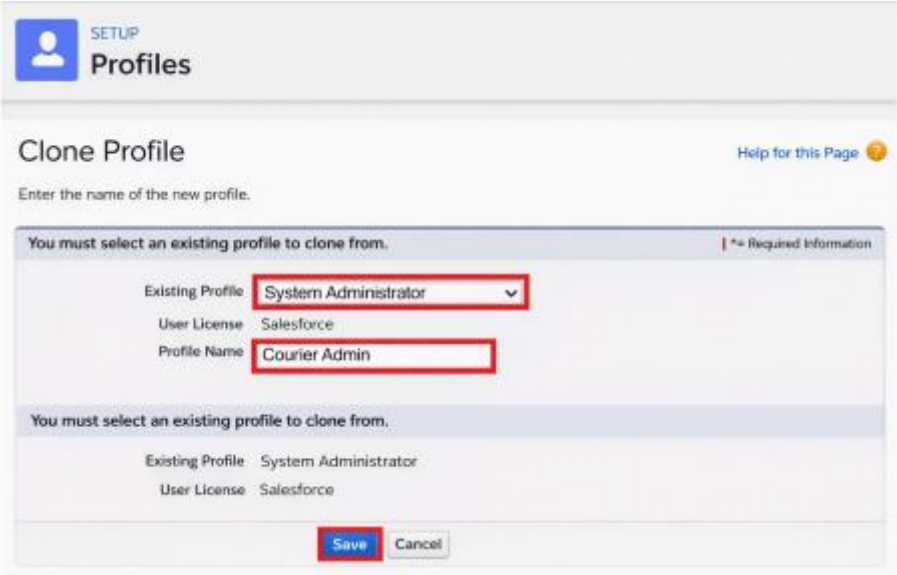
Milestone 9- Profile:

Profile

Profile is a collection of settings and permissions that control what a user can access and do in Salesforce.

Create a Custom Profile – Courier Admin :





Create a Custom Profile – Branch Manager:

Custom Object Permissions

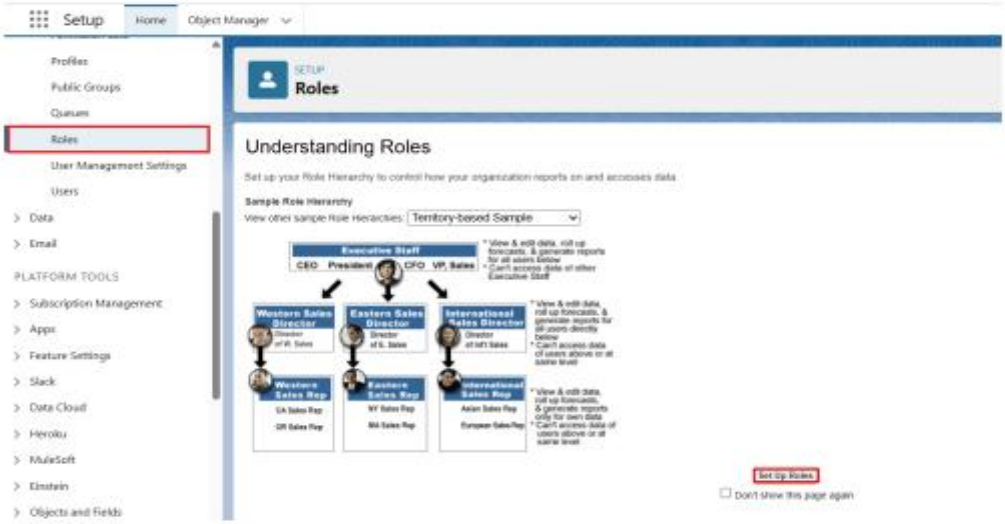
	Basic Access				Data Administration		
	Read	Create	Edit	Delete	View All Records	Modify All Records	View All Fields
Customer_c	☑	☑	☑	☑	☐	☐	☐
Shipment_c	☑	☑	☑	☑	☐	☐	☐
Delivery_Agent_c	☑	☑	☑	☐	☐	☐	☐
Branch_c	☑	☑	☑	☐	☐	☐	☐
Delivery_Update_c	☑	☑	☑	☑	☐	☐	☐
Invoice_c	☑	☐					

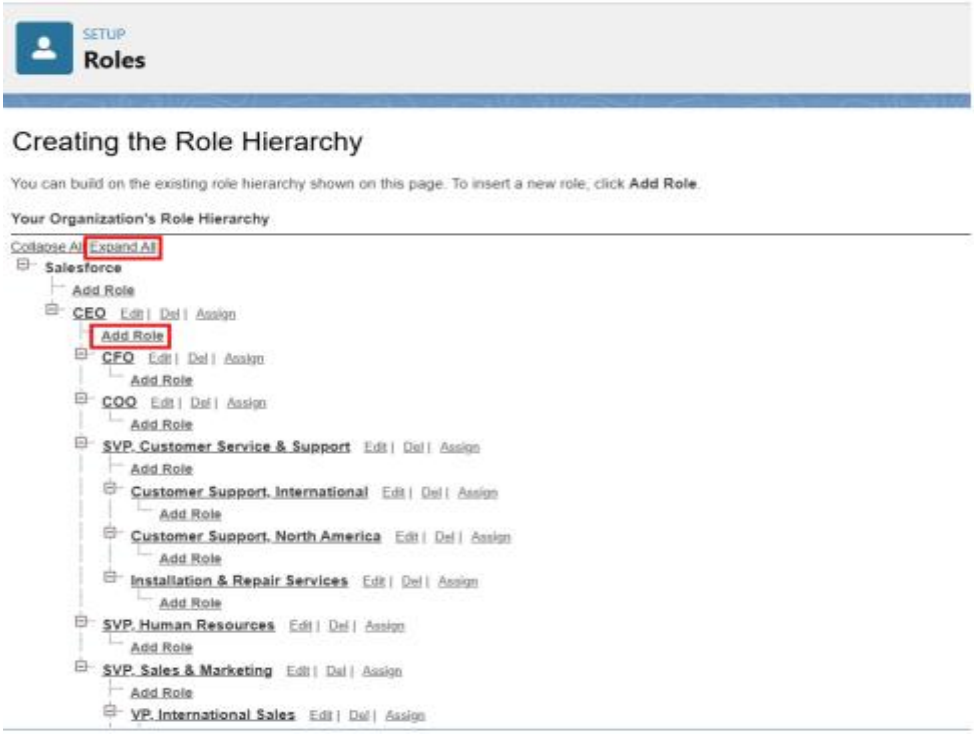
Milestone 10- Role:

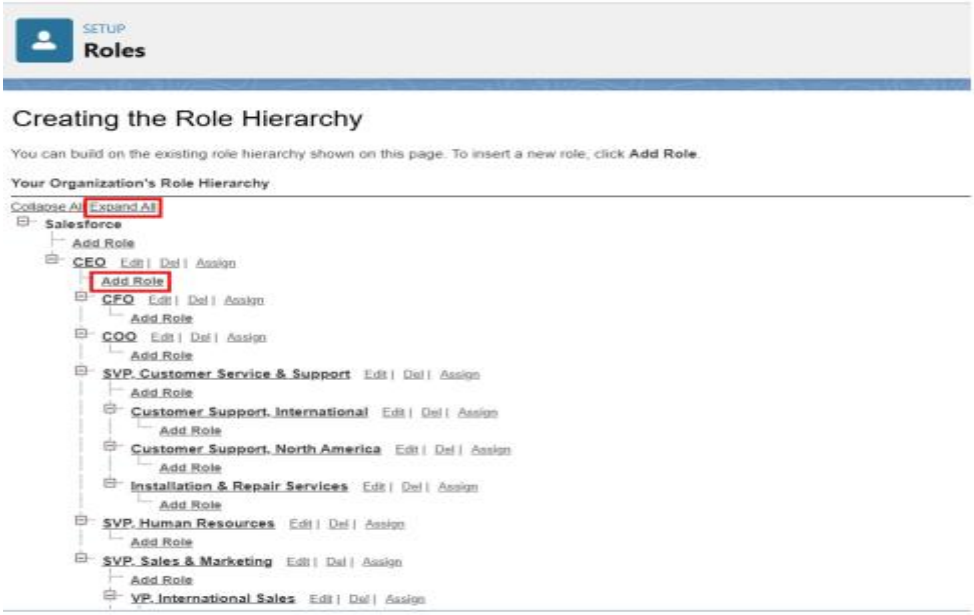
Role

Role means a user's position in the system that decides what data they can see or access.

Creation of Courier Admin Role:





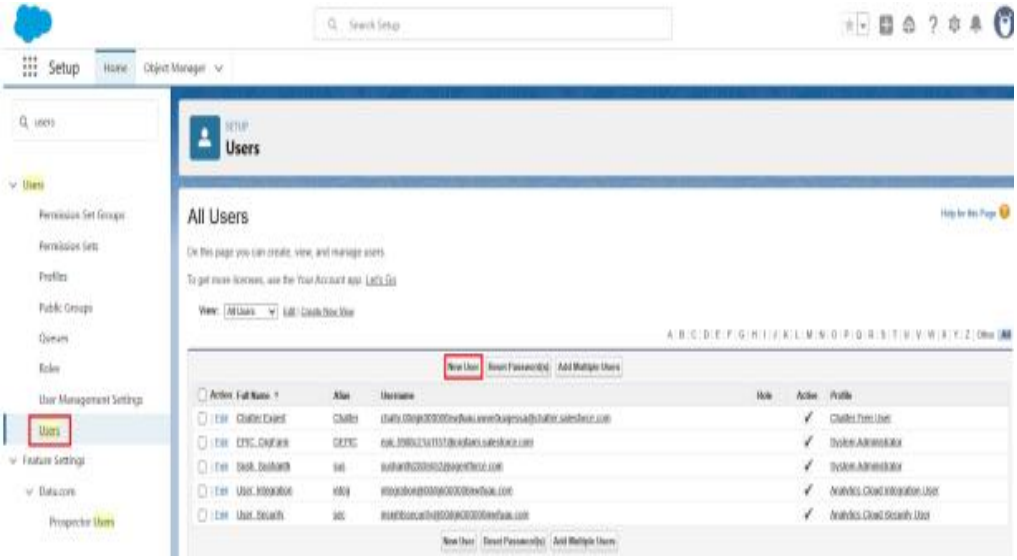


Milestone 11- User :

User

User is a person who uses a computer system or application to perform tasks.

To Create a users:



Milestone 15 : Reports & Dashboards

Reports:

Report is a list of records displayed in rows and columns used to analyze data.

Dashboards:

Dashboard is a visual display of multiple reports using charts and graphs.

Create Report Types:

Open Report Types

Create Shipment Report Type

Create Customer Report Type

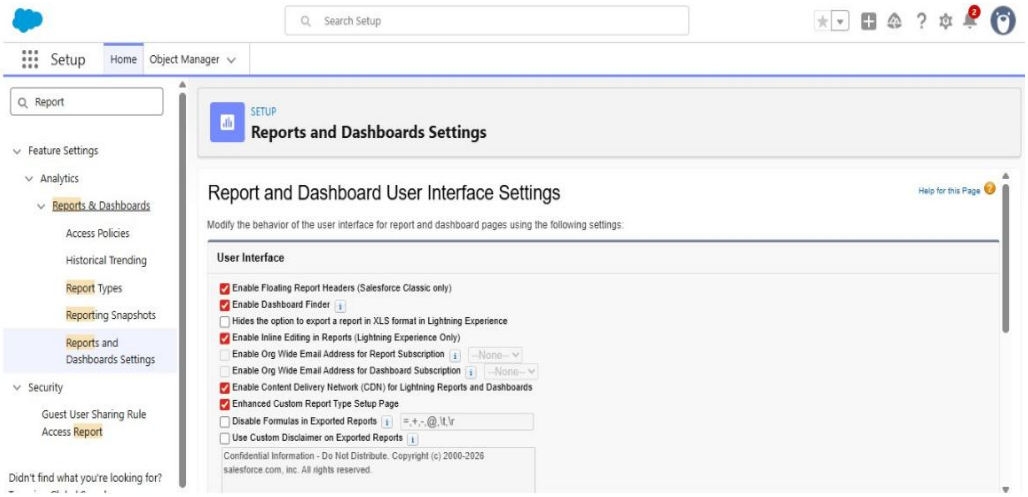
Create Agent Performance Report Type

Create Reports

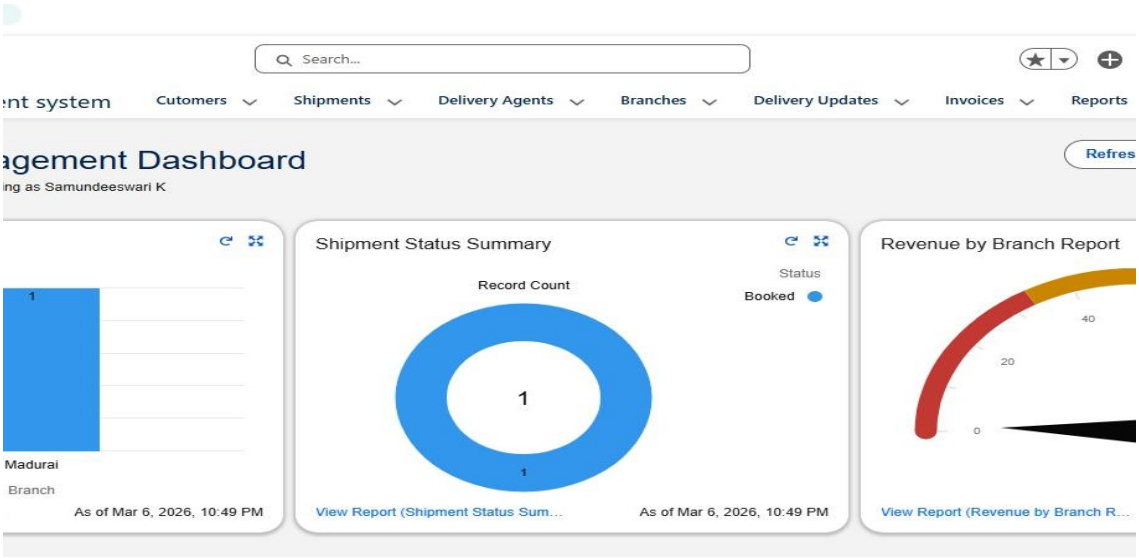
Create Dashboard

Add Dashboard Components

Save & Activate Dashboard



Final Layout:



FUNCTIONAL & PERFORMANCE TESTING:

Functional Testing is the process of testing whether all the features and functions of the system work correctly according to the requirements.

Performance testing is used to check how fast and stable the system works under different workloads

ADVANTAGES & DISADVANTAGES

Advantages:

Easy data access

Better communication

Cloud-based system

High security

Scalable platform

Realtime reporting

Disadvantages:

Internet dependency

Training required

Customization complexity

Cost for advanced features

FUTURE SCOPE

Real-time parcel tracking

Mobile app for users and delivery staff

Online payment integration

Automatic SMS/Email notifications

APPENDIX

Screenshots of the system

Sample input and output data

User manual / instructions

Reference materials used

Conclusion:

The Courier Management System helps to manage courier services easily and efficiently. It reduces manual work, improves parcel tracking, and provides better service to customers. The system makes courier operations faster, accurate, and more organized.

Thank You...